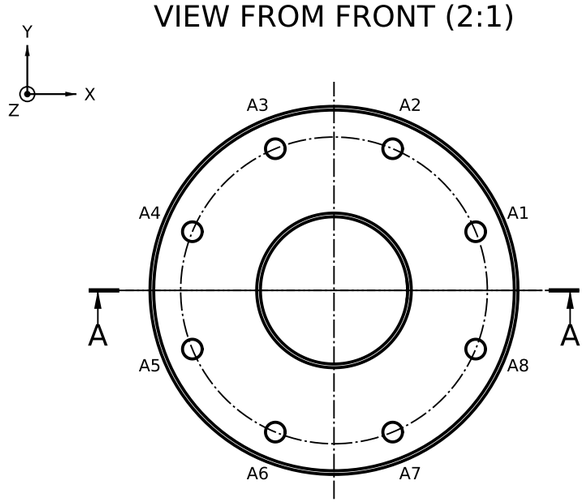
NOTES

1. Dimensions in mm. General tolerances ISO 2768-mK. Fits ISO 286-1/-2 (deviations shown in brackets).
2. STEP file governs undimensioned geometry: cnc/step/encoder_hub.step (STEVE 02e31a10 rev 48).
3. Edges ISO 13715 -0.2 (break 0.1-0.2 x 45°) unless a chamfer is shown. Remove all burrs, especially in bores and holes.
4. Surface roughness Ra 3.2 µm unless noted (ISO 21920-1).
5. Threads ISO metric coarse, 6H internal / 6g external; thread depths are full-thread depths.
6. Material AISI 303 (1.4305): non-magnetic, keep magnetic steel away from the ring's magnet. Passivate ASTM A967. (Model shows 304.)
7. Datum G: Ø12 H7 bore; datum H: ring mounting face (flange rear face, carries the RLS MRA039 ring).
8. Bore goes on the shaft's Ø12 g6 seat (AM-09 Z1) with Loctite 638. Fit with a 0.50 mm shim against the end bell rear face while it cures (sets the 2.88 ±0.13 ring-to-readhead gap).
9. Collar Ø18 must pass the readhead's Ø21.2 bore.

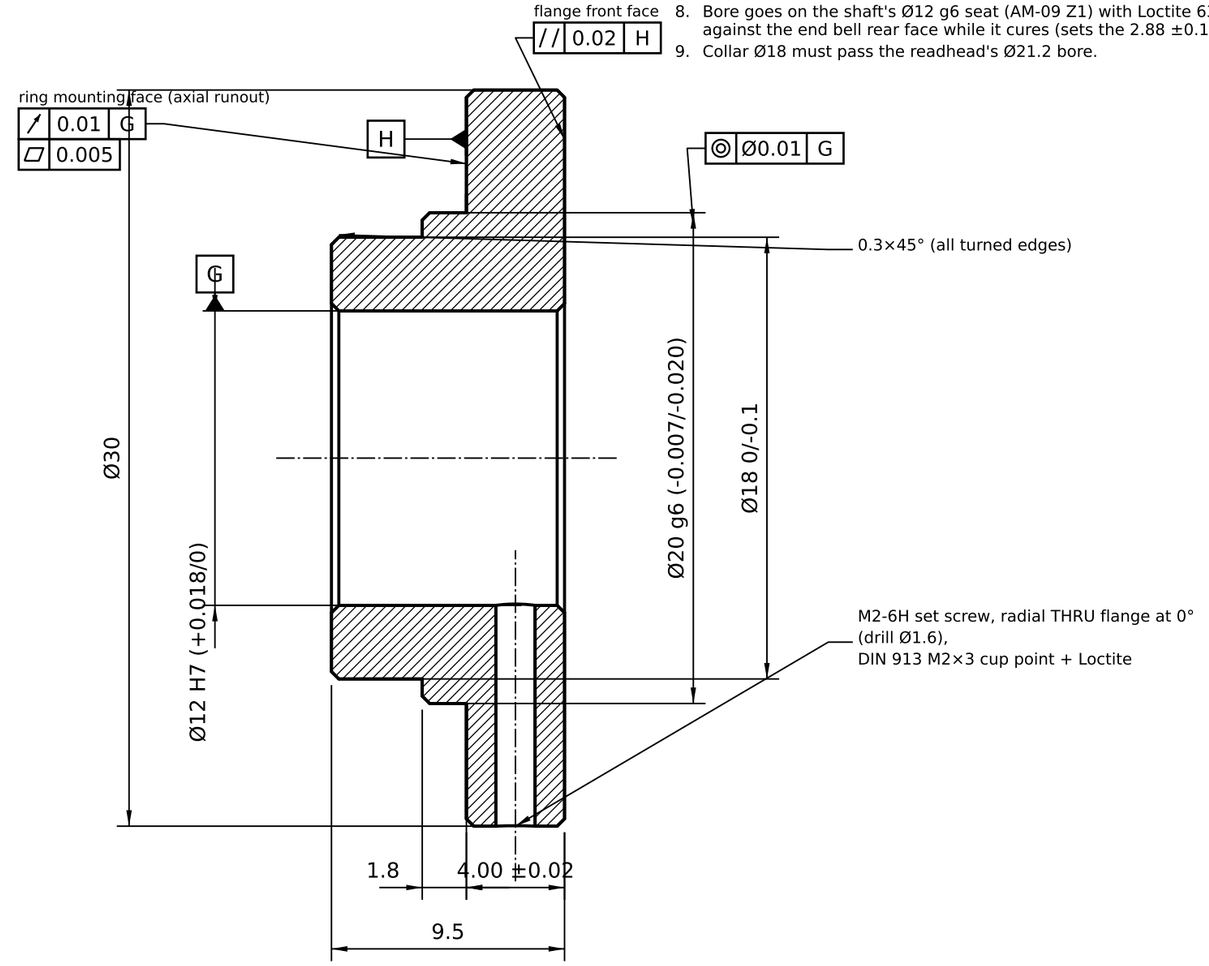
VIEW FROM FRONT (2:1)



TAG	X	Y	HOLE
A1	11.548	4.784	M2-6H THRU flange (drill Ø1.6)
A2	4.784	11.548	M2-6H THRU flange (drill Ø1.6)
A3	-4.784	11.548	M2-6H THRU flange (drill Ø1.6)
A4	-11.548	4.784	M2-6H THRU flange (drill Ø1.6)
A5	-11.548	-4.784	M2-6H THRU flange (drill Ø1.6)
A6	-4.784	-11.548	M2-6H THRU flange (drill Ø1.6)
A7	4.784	-11.548	M2-6H THRU flange (drill Ø1.6)
A8	11.548	-4.784	M2-6H THRU flange (drill Ø1.6)

1. X/Y in the STEP frame (Z axis = datum G).
2. A1-A8: position Ø0.05 to G. ISO 7046 M2×6 ring screws, 4 engaged.

SECTION A-A (4:1)



REV	DESCRIPTION	DATE	BY
A	First release for quotation and manufacture	2026-10-14	Claude

PROJECT: **Unknown machine actuator**	DRAWN BY: Claude for macncrash			SOURCE (MODEL GOVERNS): STEVE 02e31a10 rev 48 STEP: cnc/step/encoder_hub.step	
GENERAL TOLERANCES: ISO 2768-mK (ISO 22081) Edges ISO 13715 -0.2 (0.1-0.2)		PROJECTION: FIRST ANGLE ISO 128 / ISO 5456-2			
FINISH / HEAT TREATMENT: Passivate ASTM A967				UNITS: mm	
MATERIAL: AISI 303 (1.4305)				SCALE: 4:1 (views as noted)	
TITLE: **Encoder hub**				QTY PER ACTUATOR: **1**	
DRAWING / PART NUMBER: **AM-11-encoder_hub**	REV: **A**	SHEET: 1/1	SIZE: A3	DATE OF ISSUE: 2026-10-14	